

Business Insights from the Books Catalog Analysis

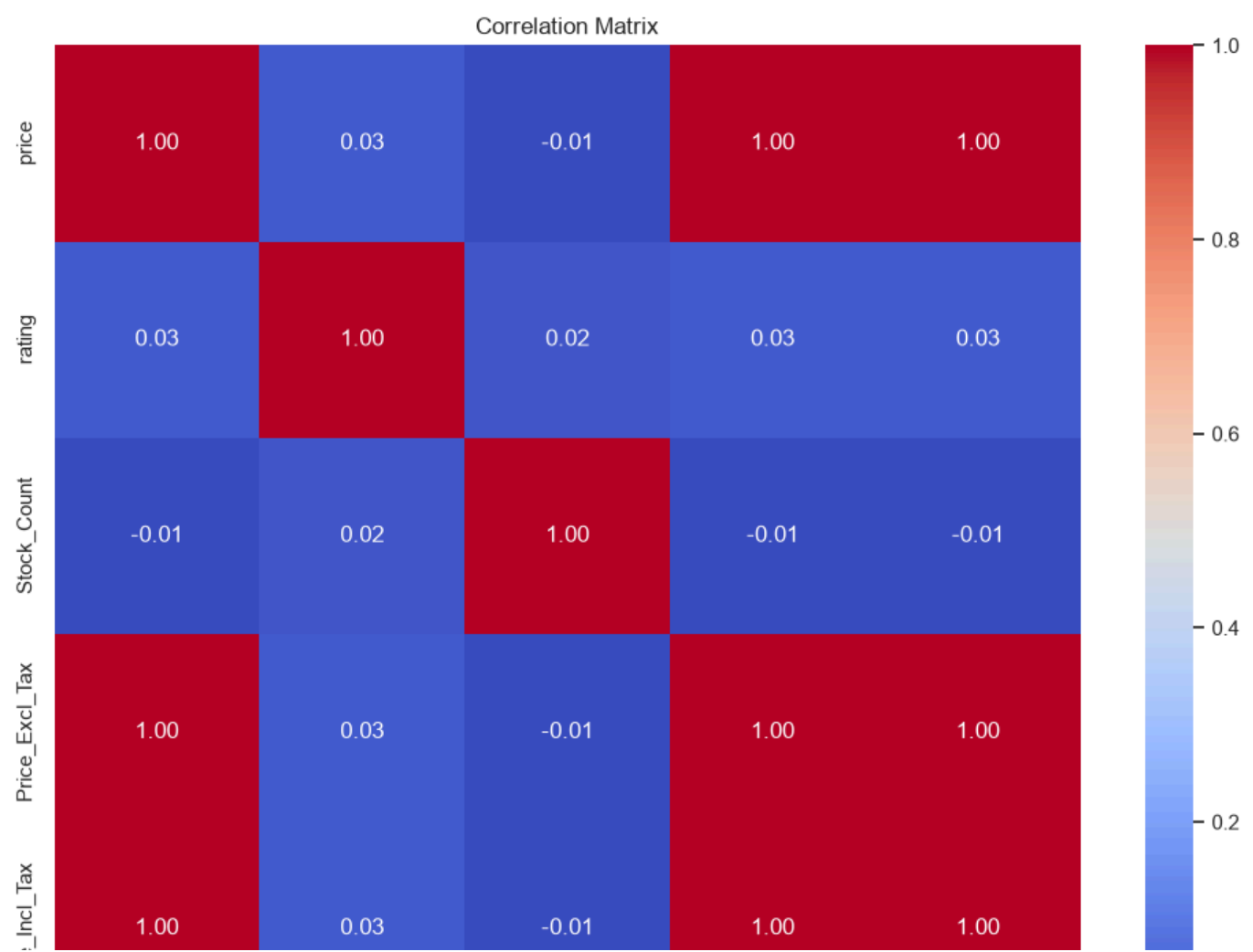
Dataset: 1,000 book records and 15 product attributes by wep

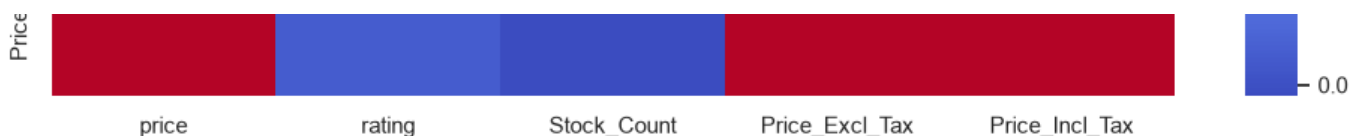
Executive summary

The analysis shows a broad book catalog with a mid-market price structure, uneven category representation, and limited evidence for demand-based decisions. The catalog has an average price of **£35.07**, a median price of **£35.98**, and prices ranging from **£10.00 to £59.99**. Inventory averages **8.585 units per book**, with stock ranging from 1 to 22 units.

The most important business caveat is that the dataset contains **zero reviews for every book, zero tax for every record, and no sales, revenue, conversion, margin, or order-date fields**. Therefore, the insights below describe catalog structure and listed inventory—not proven customer demand or best-selling performance.

1. Notebook-generated chart: Correlation matrix





Insight

The correlation matrix is mainly useful as a data-quality and modeling diagnostic. Price, price excluding tax, and price including tax are effectively the same variable in this dataset because the tax field is zero throughout. These columns should not be treated as independent commercial drivers.

The relationship between price, rating, and stock should be interpreted cautiously. Even if the visual shows weak relationships, correlation does not establish causation and cannot measure customer demand without sales or conversion data. For future analysis, the business should add units sold, revenue, gross margin, traffic, conversion rate, review count, and days in stock.

Business action

Remove redundant price fields from predictive models, validate the tax logic, and use sales and customer-behavior variables to identify the factors that actually drive performance.

2. Notebook-generated chart: Price distribution



Insight

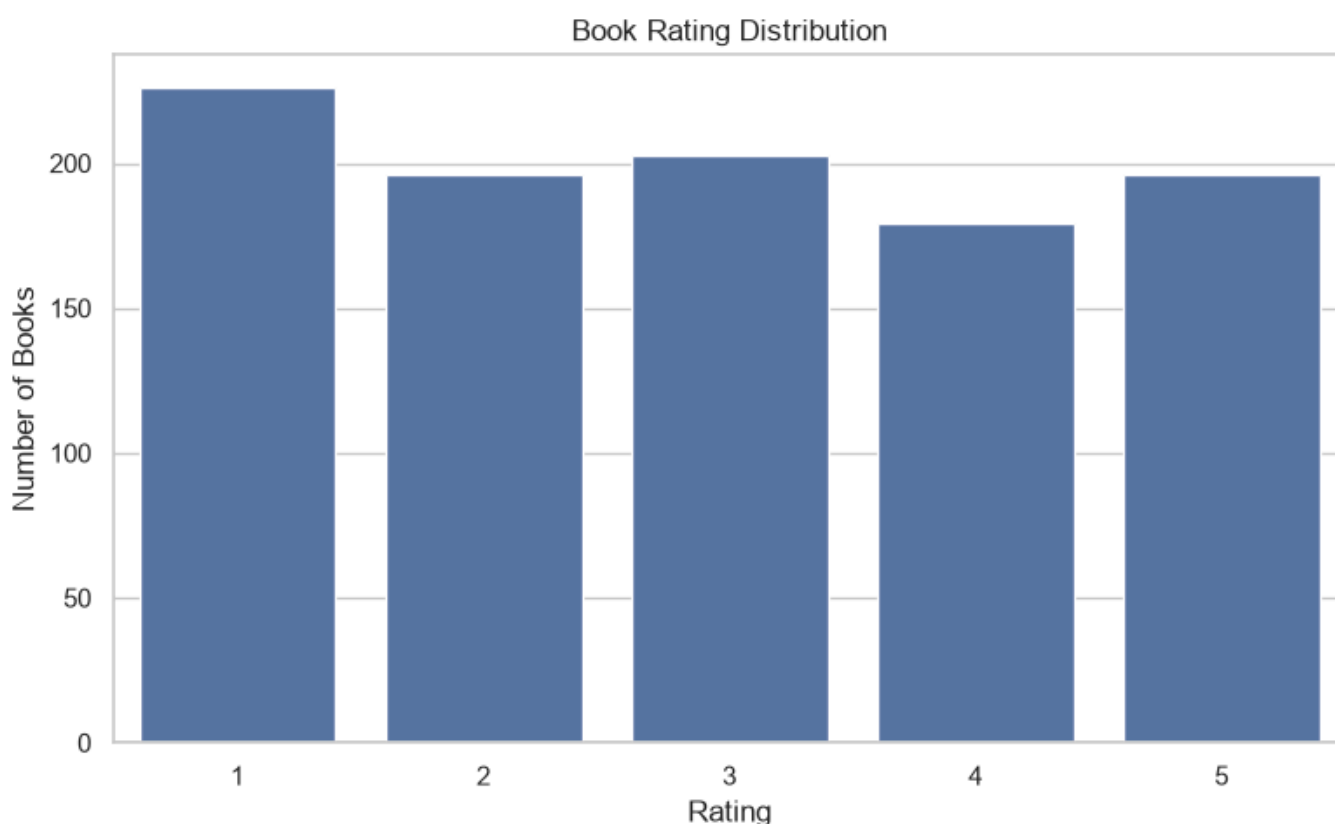
The price distribution covers a wide range from **£10 to approximately £60**, with an average of **£35.07** and a median of **£35.98**. The close proximity of the mean and median suggests that the catalog is broadly centered around a core price point rather than being dominated by a small number of extreme prices.

This range supports a practical three-tier merchandising structure: value items below roughly £22, core items between approximately £22 and £47, and premium items above approximately £47. These bands can support navigation, bundles, promotions, and gifting campaigns.

Business action

Create price-tier filters and test different promotional strategies by tier. Evaluate each tier using conversion, average order value, contribution margin, and sell-through rather than price alone.

3. Notebook-generated chart: Rating distribution



Insight

The ratings are distributed across all five levels: **226 one-star books**, **196 two-star books**, **203 three-star books**, **179 four-star books**, and **196 five-star books**. The distribution is

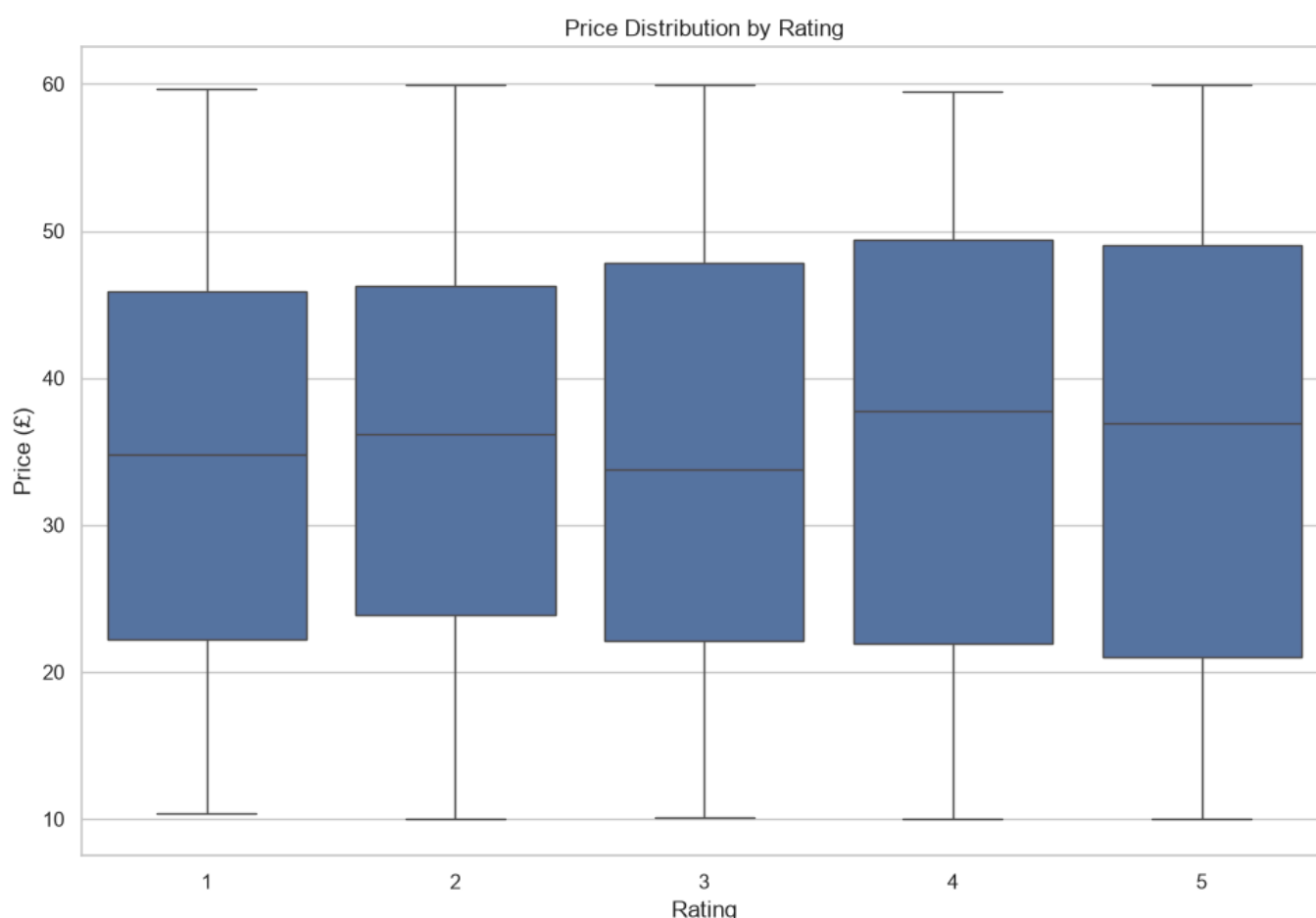
unusually balanced, and the `Number_of_Reviews` field is zero for all 1,000 records.

This means the ratings should not automatically be interpreted as reliable customer sentiment. They may represent scraped metadata, placeholder values, or an unpopulated review system. Using them alone to decide which books to discount, advertise, or reorder could lead to incorrect decisions.

Business action

Capture review count, review recency, verified-purchase status, units sold, conversion rate, and return rate. Only then should rating be used as a major merchandising or pricing signal.

4. Notebook-generated chart: Price distribution by rating



Insight

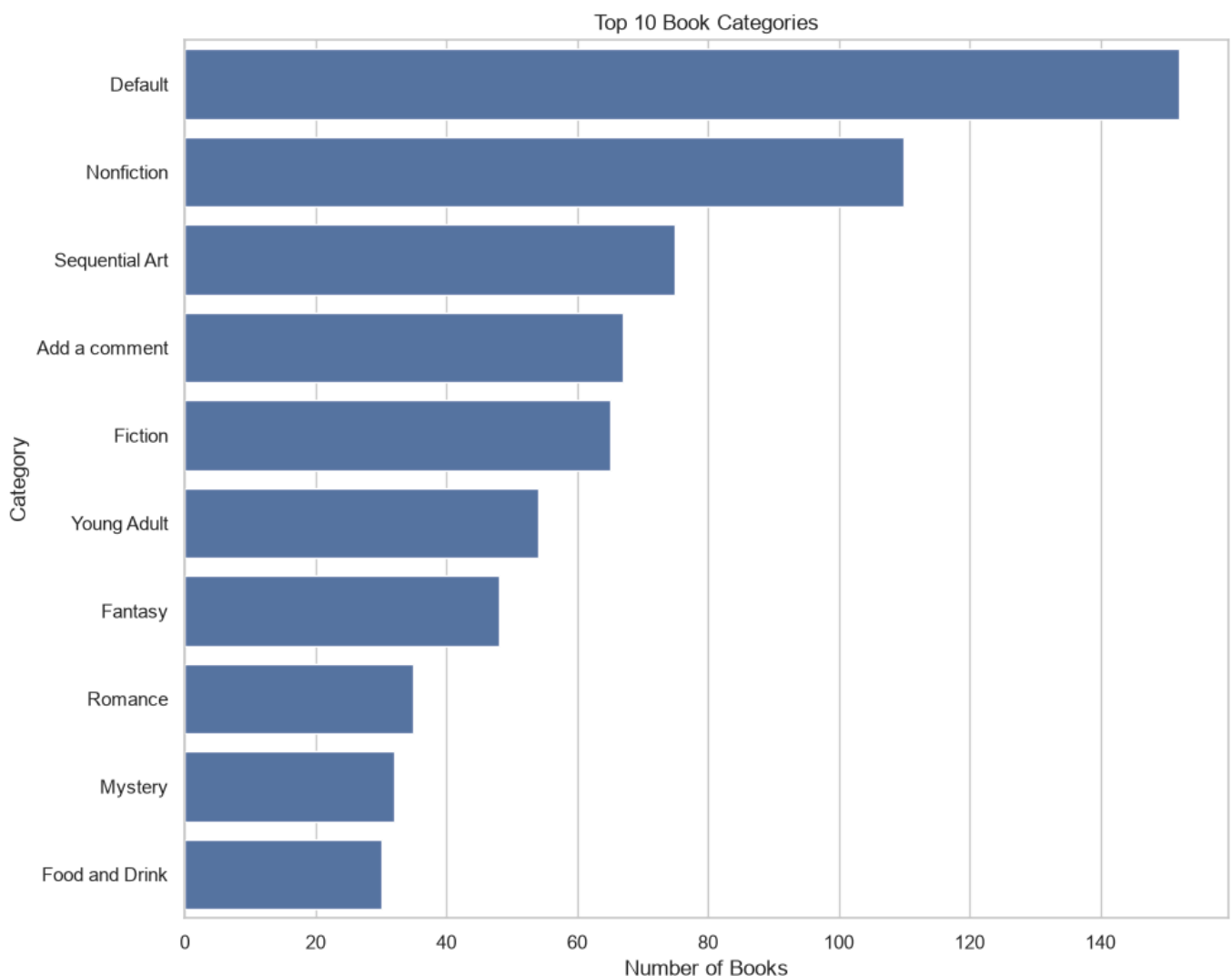
The price distributions overlap heavily across rating levels. Average prices are approximately **£34.56 for one-star books**, **£34.81 for two-star books**, **£34.69 for three-star books**, **£36.09 for four-star books**, and **£35.37 for five-star books**. The difference between the lowest and highest average is only about **£1.53**.

This suggests that price is not strongly differentiated by rating in the catalog. However, it does not prove that rating has no effect on demand because there are no sales or conversion outcomes. A high-rated book may still underperform if it receives little traffic, while a lower-rated book may sell because of brand, topic, or price.

Business action

Use controlled price and promotion experiments. Measure incremental units sold, revenue, conversion, and contribution margin by rating and price tier before changing the pricing strategy.

5. Notebook-generated chart: Top categories



Insight

The category distribution is concentrated in a few labels. The largest categories are **Default with 152 books**, **Nonfiction with 110**, **Sequential Art with 75**, **Add a comment with 67**,

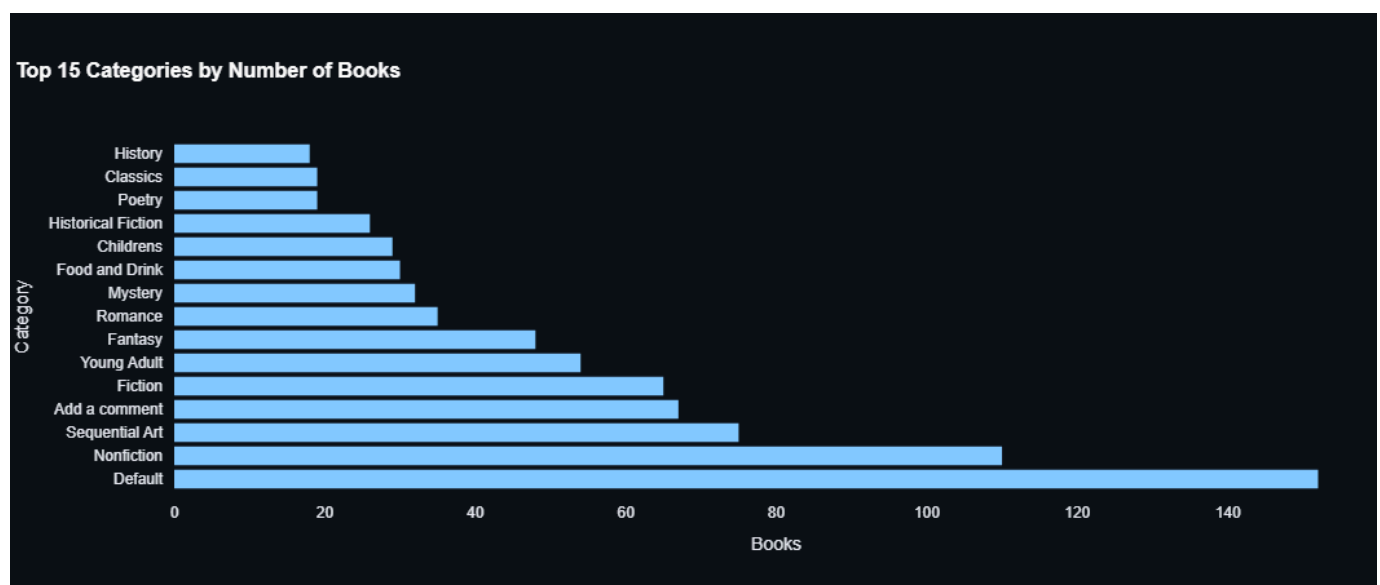
Fiction with 65, and Young Adult with 54.

The key finding is the data-quality problem: **Default** and **Add a comment** are not useful customer-facing categories and together represent **219 books, or 21.9% of the catalog**. This makes the current category ranking unsuitable for final assortment or marketing investment decisions.

Business action

Create a controlled category taxonomy and remap placeholder labels into meaningful genres. Once the cleanup is complete, build category landing pages and measure performance using traffic, conversion, revenue, margin, and inventory turnover.

6. Submitted chart: Top 15 categories by number of books



Insight

The submitted chart reinforces the concentration pattern. **Default** is substantially larger than the other categories, followed by **Nonfiction** and **Sequential Art**. This makes the catalog appear broad, but part of the apparent breadth is caused by inconsistent or placeholder category labels.

From a customer-experience perspective, the business should avoid presenting **Default** or **Add a comment** as visible shopping categories. They should be treated as records requiring classification. After correction, the business can identify which genuine genres have enough assortment depth to support dedicated navigation, campaigns, and recommendation modules.

Business action

Prioritize category cleanup before using this chart to guide buying decisions. Then compare category size with actual demand to distinguish between over-assorted categories and genuinely high-performing categories.

7. Submitted chart: Relationship between rating and price



Insight

The submitted chart shows that books across ratings from 1 to 5 appear throughout the entire price range of approximately **£10 to £60**. There is substantial vertical overlap at every rating level, so the visual does not show a clear linear relationship between rating and price.

The commercial interpretation is that a book’s listed price is not determined consistently by its rating in this catalog. This supports a test-and-learn approach rather than a rule such as “high-rated books should always be more expensive.” The result is also consistent with the notebook’s grouped averages, which are close across all rating levels.

Business action

Do not use rating alone for repricing. Combine rating with review volume, sales velocity, conversion, competitive price, stock age, and gross margin. This will allow the business to separate genuine customer preference from incomplete or unreliable catalog metadata.

Management priorities

Priority	Recommendation	Expected business benefit
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1	Clean Default and Add a comment category labels	Better navigation, reporting, and assortment decisions
2	Add sales, revenue, margin, conversion, and review data	Enables demand-based pricing and purchasing decisions
3	Create value, core, and premium price tiers	Improves merchandising and promotion design
4	Monitor books with only 1–3 units in stock	Reduces stockout exposure
5	Test prices and promotions experimentally	Measures actual customer response instead of assuming it
6	Validate tax and duplicate price fields	Prevents incorrect financial and analytical conclusions

Final conclusion

The strongest immediate opportunity is to improve **catalog structure and data quality**. The category charts reveal that more than one-fifth of the records sit in placeholder or invalid labels, while the rating-price charts show no clear relationship between listed price and rating. The business should therefore treat the current analysis as a catalog baseline and build a second version using transaction and customer-behavior data before making major pricing, purchasing, or marketing decisions.